

STATEMENT OF WORK (SOW)

Scientific Evidence Evaluation and Knowledge Integration for Translational Toxicology

April 13, 2026

1 General Information

1.1 General

This is a non-personal service(s) contract under which the personnel rendering the service(s) are not subject, either by the contract's terms or by the manner of its administration, to the supervision and control usually prevailing in relationships between the Government and its employees. The purpose of this contract is to provide all services and supplies necessary to conduct scientific evidence evaluation and knowledge integration for translational toxicology in support of the National Institute of Environmental Health Sciences (NIEHS), Division of Translational Toxicology (DTT).

1.2 Description of Services

Independently, and not as an agent of the Government, the Contractor shall furnish all the necessary services, qualified personnel, procedures, materials, equipment, and facilities not otherwise provided by the Government as needed to perform the objectives set forth below. In general, activities shall be conducted according to the requirements described in a research outline (see Section 3.1), which includes project-specific activities, milestones, and deliverables.

1.3 Background and Scope

DTT is an intramural division of NIEHS, National Institutes of Health, that uses traditional and cutting-edge research approaches to better understand how factors in our environment may impact human health. Such approaches include using scientific expertise to identify, collect, analyze, and interpret diverse types of information relevant to the effects of environmental exposures on human health. Information from the fields of toxicology, epidemiology, chemistry, data science, informatics, pathology, and other disciplines is used in the design of research programs; in literature-based health assessments; and in the development of novel tools and informatics approaches.

The objective of this contract is to provide scientific and technical expertise and support for scientific evidence evaluation and knowledge integration for translational toxicology related to the evaluation and reporting of effects of environmental substances and other issues that may impact public health. Nine specific requirement areas under this contract to support DTT within NIEHS include: (1) systematic evidence mapping and scoping reviews of environmental health literature, (2) systematic review and evidence synthesis, (3) report development and scientific writing, (4) document preparation and publication, (5) coordination and management of external peer review, (6) logistical management for advisory group meetings and workshops, (7) provision of scientific or technical consultants with expertise in relevant scientific disciplines, (8) development and implementation of novel tools and informatics approaches, and (9)

program management. These requirement areas are managed by several offices and branches within DTT. Additional information about the DTT organizational structure is available on the [NIEHS website](#).

The National Toxicology Program (NTP) is an interagency partnership comprised of NIEHS (part of the National Institutes of Health), the National Institute for Occupational Safety and Health (NIOSH; part of the Centers for Disease Control and Prevention), and the U.S. Food and Drug Administration (FDA; primarily at the National Center for Toxicological Research). NTP is headquartered administratively at NIEHS, and many of the activities performed under this contract will support NTP projects and research programs.

The activities carried out by this contract will address the aforementioned requirements for scientific evidence evaluation and knowledge integration for translational toxicology.

1.4 Overview of Required Capabilities

The Contractor shall conduct activities supported by the following Capability Areas described in more detail in Section 2 – Required Capability Areas:

- Systematic Evidence Mapping and Scoping Reviews of Environmental Health Literature
- Systematic Review and Evidence Synthesis
- Report Development and Scientific Writing
- Document Preparation and Publication
- Coordination and Management of External Peer Review
- Logistical Management for Advisory Group Meetings and Workshops
- Provision of Scientific or Technical Consultants with Expertise in Relevant Scientific Disciplines
- Development and Implementation of Novel Tools and Informatics Approaches
- Program Management

1.5 Policies and Procedures

For scientific evidence evaluation and knowledge integration for translational toxicology, the Contractor and applicable subcontractor(s) shall comply with the below guidelines:

- a. Handbook for Conducting Systematic Reviews for Health Effects Evaluations
 - Available at: <https://ntp.niehs.nih.gov/go/ohathandbook>.
- b. Handbook for Preparing Report on Carcinogens Monographs
 - Available at: <https://ntp.niehs.nih.gov/go/rochandbook>.

1.6 Reports and Deliverables

Information on specific reporting requirements is presented below. All deliverables shall be electronic and PC and MAC compatible unless otherwise specified by the Contracting Officer's Representative (COR). Each final document or report shall be submitted electronically in MS Word and/or PDF formats to the COR for review. Unless otherwise directed, all submitted final electronic deliverables shall be compliant with Section 508 regarding accessibility (see Section 1.6.4). In some cases, there may be accommodations for which the Government will provide information annually. Unless otherwise instructed, all reports submitted under this contract

shall reference the contract number, the title and type of report, and the submission date on the first page of the report.

In addition to the reports mentioned herein, any incidence or unforeseeable occurrence not listed here, including personnel changes which may impact activities under the contract, shall be immediately reported to the Government.

DELIVERABLE	REFERENCE	DUE/FREQUENCY	SUBMIT TO
Quality Management Plan	Section 1.6.1.1	Revise as needed or when major changes occur	COR, CO, and CS
Monthly Progress Report	Section 1.6.1.2	Due once per month	COR, CO, and CS
Monthly Budget Summary Report	Section 1.6.1.3	Due once per month as an attachment with the monthly invoices	COR, CO, and CS
Annual Report	Section 1.6.1.4	Due 4 weeks from the last day of each contract year	COR, CO, and CS

COR = Contracting Officer's Representative; CO = Contracting Officer; CS = Contract Specialist.

1.6.1 Periodic Reports

1.6.1.1 *Quality Management Plan*

A Quality Management Plan (QMP) shall be submitted before performance of work under the contract and updated versions submitted at regular intervals (e.g., biannually).

1.6.1.2 *Monthly Progress Report*

Once per month, the COR shall receive a summary of the work currently in progress. This report shall include details on the status of all active work, current costs for active work, cumulative costs for the contract (including direct and administrative costs), any problems encountered, and remedial actions taken to resolve those problems. The monthly progress report shall also include a summary of the work performed by any subcontractor and/or ad hoc consultant, including their deliverables, hours spent, and cost. This summary shall also include a description of how the subcontractor/consultant work aided the Contractor in completing or nearing completion of a specific project.

1.6.1.3 *Monthly Budget Summary Report*

A summary report shall be submitted monthly that includes budget information for each activity separately undertaken and for the contract as a whole. The report must include each activity's budget, total expenditures on the activity for the month, summaries of current contract year and cumulative expenditures, and projected expenses to complete each activity. Budget summary reports shall be submitted monthly to the COR as MS Excel spreadsheets (or in other spreadsheet software by agreement) and shall be sortable as required by the COR.

Budget summary reports shall be included as an attachment with the monthly invoices.

1.6.1.4 Annual Report

An annual report shall be submitted within 4 weeks of the last day of each contract year. Each annual report shall detail all significant outcomes resulting from the requirements as described within the SOW and contract, including a breakdown of costs by research activity incurred to date, any problems encountered and remedial actions taken to resolve the problem, and any deviations from the SOW and/or contract.

1.6.2 Research Products

For each activity performed under the contract, research products shall be generated by the Contractor and submitted to the COR according to the milestones and deliverables schedule included in the research outline. The type and number of product(s) may vary depending on the nature of the activity.

1.6.3 IT Security Plan and Other IT Deliverables

Refer to Appendix A for specific requirements related to information technology (IT) security.

1.6.4 HHS Section 508 and Accessibility Standards

Section 508 of the Rehabilitation Act, as amended by the Workforce Investment Act of 1998 (P.L. 105-220) requires that when Federal agencies develop, procure, maintain, or use information and communication technology (ICT), it shall be accessible to people with disabilities. Federal employees and members of the public who have disabilities must have access to, and use of, information and data that is comparable to people without disabilities.

All products, platforms and services delivered as part of this work statement that are ICT, or contain ICT, must conform to the Revised 508 Standards, which are located at 36 C.F.R. § 1194.1 & Apps. A, B, & C, and available at <https://www.access-board.gov/ict/>.

All requirements are applicable to support services and documentation deliverables. All functional performance criteria apply when using an alternative design or technology that achieves substantially equivalent or greater accessibility and usability by individuals with disabilities, than would be provided by conformance to one or more of the requirements in Chapters 4-6 of the Revised 508 Standards, or when Chapters 4-6 do not address one or more functions of ICT.

For each proposed product or platform, a fully completed Accessibility Conformance Report (ACR) using the Voluntary Product Accessibility Template (VPAT) (<https://www.itic.org/policy/accessibility/vpat>) or HHS conformance checklist(s) (<https://www.hhs.gov/web/section-508/accessibility-checklists/index.html>) must be submitted.

Prior to acceptance of deliverables, the offeror must demonstrate conformance to the HHS Section 508 requirements via HHS Section 508 checklist(s) (<https://www.hhs.gov/web/section-508/accessibility-checklists/index.html>). The Government reserves the right to perform testing on required ICT items to validate the offeror's Section 508 conformance claims. If the Government determines that Section 508 conformance does not meet the HHS Section 508 requirements, the Government shall, at its option, require the offeror to remediate the item, at no additional cost to the Government, to align with the HHS Section 508 conformance requirements prior to acceptance.

References:

- Section 508 Standards: <https://www.access-board.gov/ict/>
- HHS Policy on Section 508 Compliance and Accessibility of Information and Communications Technology (ICT): <https://www.hhs.gov/web/governance/digital-strategy/it-policy-archive/hhs-policy-section-508-compliance-accessibility-information-communications-technology.html>
- HHS Accessibility and Section 508 Conformance Checklists: <https://www.hhs.gov/web/section-508/accessibility-checklists/index.html>

1.7 Post Award Conference/Periodic Progress Meetings

The Contractor shall attend any post award conferences convened by the contracting activity or contract administration office in accordance with FAR Subpart 42.5. The Contracting Officer (CO), COR, and other Government personnel, as appropriate, may meet periodically with the Contractor to review the Contractor's performance. At these meetings the CO will apprise the Contractor of how the Government views the Contractor's performance and the Contractor shall apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

1.7.1 Progress Meetings

The Contractor shall attend, participate in, and furnish input to scheduled and unscheduled meetings, conferences, and briefings that relate to the functions and services herein as required by the Government to provide effective communication and impart necessary information. The Principal Investigator (PI), Program Manager (PM), and/or designated representative shall attend meetings as requested by the Government. Meeting attendees shall at times include Contractor managerial, supervisory, and other personnel knowledgeable of the subject matter. Meetings may start or end outside of regular duty hours.

1.8 Contracting Officer's Representative

As determined by the CO, a COR will be appointed and identified by letter of designation, a copy of which will be provided to the Contractor by the CO. The designation letter states the responsibilities and limitations of the COR. The COR has the responsibility and authority to monitor all aspects of the day-to-day administration of the contract. Responsibilities include ensuring the timeliness and delivery of quality products and services to meet the requirements of the contract.

1.9 Personnel

This contract requires a high degree of collaboration among (or by) Contractor staff. Each requirement described in this SOW shall be performed by an appropriate mix of professional and technical staff. The Contractor shall employ and/or have access to staff with broad professional and technical expertise in environmental health sciences necessary to conduct evaluations of complex scientific literature and fulfill the activities of the SOW. Contract personnel shall have the expertise to contribute to problem formulation and research design as required. Contractor staff may need to be available after normal working hours and on weekends.

Personnel assigned to this contract, including subcontractors and/or consultants, shall demonstrate to the COR that there are no conflicts of interest relative to the work to be conducted prior to performing work under the contract. Personnel assigned to this contract with access to federal IT resources may be required to pass a security investigation (see Section 4).

The Contractor shall ensure that all Contractor personnel are adequately trained, possess the required credentials, and are fully qualified and skilled to perform the activities outlined in this SOW. The Contractor shall be responsible for all costs associated with technical training and necessary travel to ensure that staff maintain required certifications and proficiency to perform assigned duties.

1.9.1 Professional Personnel

Contractor staff designated as key personnel must be approved by the CO and the COR and shall be employees of the prime Contract organization and members of the professional staff.

The PI and the PM are both considered key personnel. The PI and the PM shall have appropriate training, experience, and scientific expertise and shall be familiar with the management and coordination of multidisciplinary projects relevant to the requirements for scientific evidence evaluation and knowledge integration for translational toxicology described in the SOW.

Senior toxicologists, senior epidemiologists, staff toxicologists, and staff epidemiologists are considered professional staff due to the nature of the data used and generated by DTT, and these staff shall be able to use independent judgment based upon their scientific expertise. The daily interaction and constant coordination of efforts needed among these disciplines requires that they be organizationally together.

Senior toxicologists shall have appropriate education, training, and experience in (1) the conduct and/or evaluation of studies in toxicology, carcinogenicity, reproductive and developmental biology, or a related field; (2) evaluating study quality/informativeness, interpreting evidence from individual studies, and integrating evidence across a collection of relevant studies; (3) extracting, visualizing, and/or analyzing data; and (4) working collaboratively and preparing scientific reports.

Senior epidemiologists shall have appropriate education, training, and experience in (1) the conduct and/or evaluation of environmental and occupational observational epidemiology studies in health-related fields; (2) evaluating study quality/informativeness, interpreting evidence from individual studies, and integrating evidence across a collection of relevant studies; (3) extracting, visualizing, and/or analyzing data; and (4) working collaboratively and preparing scientific reports.

Staff toxicologists and staff epidemiologists shall have appropriate training and experience in conducting or preparing scientific reviews of toxicological or observational epidemiological studies, including extracting data from studies; evaluating study quality/informativeness; interpreting and summarizing study findings; and extracting, visualizing, and/or analyzing data.

1.9.2 Technical Personnel

Scientific expertise may be required in chemistry, toxicology, environmental and occupational epidemiology, cancer epidemiology, statistics, mechanisms of effect, in vitro analyses, computational biology, data science, informatics, environmental exposures (e.g., occupational hygienists, exposure science), read-across, genotoxicity, disposition and metabolism, carcinogenicity, immunotoxicology, reproduction, development, or other areas relevant to activities conducted under the contract. In addition, supporting staff are required for the following areas: scientific writing and editing, data visualization, and meeting logistics. The level of effort for each discipline, at any time, may vary from year to year based on the complexity, nature, and extent of the assigned work. When experts who are not part of the Contractor's staff are needed, the Contractor shall have procedures in place to ensure the needed expertise can be obtained and provide the COR within one week a plan for obtaining that expertise.

1.10 Data Rights, Patents, and Copyrights

The Government has unlimited rights to all documents and materials produced under this contract. All documents and materials, including the source codes of software, produced under this contract shall be Government owned and are the property of the Government with all rights and privileges of ownership and copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the CO. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. This right does not abrogate any other Government rights.

1.10.1 Source Documentation

All original source documentation is the property of NIEHS. Data summaries and other relevant information generated during the performance of this contract shall be submitted to the COR in the form of status reports and agent specific reports as specified in the Deliverables section above. In addition, annual reports shall be submitted for all activities as specified in the Deliverables section below. To the extent possible, the Government will retain unrestricted access and rights to all data and technical information generated under the contract.

1.11 Non-Disclosure Requirements

Performance under this contract may require the Contractor to access data and information proprietary to a Governmental agency, another Government Contractor, or of such nature that its dissemination or use other than as specified in this work statement would be adverse to the interests of the Government or others. The Contractor and Contractor's employees (including Subcontractors) shall not divulge or release data or information developed or obtained under performance of this contract, except to authorized Government personnel or upon written approval of the CO. The Contractor shall not use, disclose, or reproduce proprietary data, which bears a restrictive legend, other than as specified in this contract. All documentation showing individual names or other personal information shall be controlled and protected under the provisions of the Privacy Act of 1974, Public Law 93-579, 5 United States Code (U.S.C.) Section 552a.

1.12 Replication, Dissemination, or Use of the Results

It is anticipated that the results of the research conducted under the contract shall be published in the open, peer-reviewed scientific literature and/or in an NTP, NIEHS, or DTT report. Specific details concerning authorship and technical issues surrounding the publication of the research shall be discussed and agreed upon by the PI and the COR on a case-by-case basis.

Because premature release of information developed under the contract could be harmful to the public, the Contractor and the Government agree that it is important to consider issues related to the timing of the release of research results when submitting information for publication. Regarding public disclosures of the research to be carried out pursuant to this contract, the Contractor agrees to submit all proposed manuscripts, abstracts, or reports to the COR at least thirty (30) days in advance of submission for publication. Within this thirty-day period the COR shall inform the Contractor if there are serious concerns raised by the timing of said publication. Upon receipt of a notification that the Government objects to the publication of said information, the Contractor agrees to withhold submission of said publication while the author and the COR discuss the issues of concern. The Contractor agrees to give good faith consideration to all such issues and further agrees that it will not knowingly submit for publication any article, such as a manuscript, abstract, or other item, that would be harmful to the public welfare. The final decision on whether to submit an article for publication shall rest with the COR after the procedures set forth above are followed. Similar procedures shall apply for scientific presentations and meeting abstracts.

1.13 Protection of Government and Contract Information

The Contractor shall not cite any information (e.g., contract information, pictures, locations, etc.) obtained through this contract on any hard copy or digital marketing tools to include its company website.

1.14 Non-Disclosure Statements

The Contractor shall provide signed non-disclosure agreements to the Government no later than five (5) days prior to commencement of work under the contract. Disclosure of information by Contractor personnel may result in removal of Contractor personnel from performance under this contract.

1.15 Safety Laws and Regulations

The Contractor shall comply with FAR 52.236-13, Accident Prevention (November 1991).

In addition to other requirements for accident prevention specified in this contract, the Contractor shall comply with all Federal and State requirements governing safety and health protection.

The Contractor shall indicate its plan for compliance considering conditions of the work described by the design, technical proposal, operational activities of the Government or others in the area of the work, identification of the coordinator, and implementation of all other accident prevention provisions of this contract.

The Contractor shall complete and provide safety notifications as required by national, state, and local regulatory bodies, and will provide documented proof to the CO via the COR, no later than 10 days that such notifications were made.

2 Required Capability Areas

2.1 General Requirements

The Contractor shall provide services, qualified personnel, equipment, and facilities needed to support scientific evidence evaluation and knowledge integration for translational toxicology under this contract. For all projects under the contract, the Contractor shall compile all working files and document all relevant correspondence and decisions for project files to be submitted to the COR after project completion. Documentation shall include relevant correspondence both with the Government and internal to the Contractor. A project file acts as the official repository for documenting the entire research process, and is an organized, comprehensive collection of records, documents, and data pertaining to a specific research project. The components of a project file may differ by project type.

2.2 Systematic Evidence Mapping and Scoping Reviews of Environmental Health Literature

The Contractor shall provide staff with expertise in literature searching, literature evaluations, and the management of systematic evidence mapping and scoping reviews. Domain-specific expertise for this capability area shall include toxicology and subdisciplines (e.g., reproductive toxicology, cancer), epidemiology, chemical or biological properties, human exposure, mechanistic data, and informatics. Staff shall have access to a variety of specialized databases and information resources related to environmental health.

Most activities conducted under this capability area will require support as outlined in Section 2.2.1; however, in some cases, Contractor support for a more limited scoping review will be required to address a given research question or develop a research concept. For example, brief or targeted scoping reviews may be part of problem formulation to refine the exposure, outcomes, or animal models of interest.

2.2.1 Systematic Evidence Mapping

The Contractor shall provide staff with the appropriate expertise to manage and support the development of systematic evidence maps (SEMs) including literature searching, literature screening, data extraction, data categorization, and data visualization. All systematic evidence mapping activities shall be conducted according to the first three steps outlined in the Handbook for Conducting Systematic Reviews for Health Effects Evaluations¹ (problem formulation, search and selection of studies for inclusion, data extraction) with procedures adjusted to address a given topic.

2.2.1.1 Literature Searching

Literature searching to support systematic evidence mapping shall be conducted with the rigor required for systematic review. To identify relevant scientific literature for a SEM, the Contractor shall develop focused and/or comprehensive literature or information search strategies in consultation with the COR. Literature or information searches shall be conducted using multiple scientific literature databases and other relevant sources (e.g., websites, exposure and authoritative sources) and literature or data retrieved from these resources shall

¹ Available at: <https://ntp.niehs.nih.gov/go/ohathandbook>.

be organized using reference management software, preferably EndNote™. The Contractor shall also provide a mechanism for storage and management of all electronic data or literature files (e.g., file management software such as SharePoint/OneDrive).

As needed, the Contractor shall conduct searches of relevant literature databases for information on a particular agent (chemical, physical, biological) or stressor and reported associated health effects. In addition, information on use, occurrence, human exposure, and guidelines and regulations for substances under evaluation may also be collected.

Occasionally, searches of databases not available through the NIEHS or NIH library may be needed, and this will be determined on a case-by-case basis after discussions with the COR. In addition, translation of studies published in foreign journals may be needed, and this will be determined on a case-by-case basis after discussions with the COR. The Contractor shall arrange for translation services as necessary.

The Contractor shall make accessible to Government staff electronic, word-searchable, portable document format (PDF) files for all references cited in all drafts of scoping documents (see Section 2.2.3). As directed, the Contractor shall make the references electronically available, either in a transmittable EndNote file or via a shared online location (i.e., SharePoint, Teams, OneDrive), depending upon instructions from the COR. The Contractor shall provide a mechanism for file sharing of materials and references among designated Government staff or consultants (e.g., using project management software).

The Contractor shall arrange for and/or acquire any needed information resources or information management software following discussion and approval by the COR.

2.2.1.2 Literature Screening

Literature screening to support systematic evidence mapping shall be conducted with the rigor required for systematic review. The Contractor shall use web-based, secure literature screening software appropriate for supporting the transparency and reproducibility of systematic review methods for all literature screening activities. The Government will provide access to and/or reimbursement for all required software. Software examples include, but are not limited to:

- DistillerSR [<https://www.distillersr.com/products/distillersr-systematic-review-software>]
- Health Assessment Workspace Collaborative (HAWC) [<https://hawcproject.org/>]
- SWIFT Active Screener [<https://www.sciome.com/swift-activescreener/>]
- Table Builder [<https://ehp.niehs.nih.gov/doi/full/10.1289/EHP4224>]

The Contractor shall screen relevant literature using state-of-the-art approaches including text mining, machine learning, artificial intelligence (AI)-enabled, and/or manual screening; select literature using specific inclusion/exclusion criteria; and track the output of literature screening in detail for display or data storage. Studies identified as relevant will be broadly categorized into an appropriate evidence stream or selected scientific topic (e.g., human or experimental animal studies, exposure data, other relevant data including in vitro or mechanistic) as defined by the scope of the evaluation and input from the Government.

2.2.1.3 *Scientific Data Extraction, Categorization, Management, and Visualization*

The Contractor shall extract specified data from relevant scientific studies, as defined by the Government, into databases (e.g., HAWC or Table Builder) using specific web-based forms which are easily exportable to Excel and Word tables. Data shall be extracted using tiered, hierarchical methods that support interactive evidence mapping.

The Contractor shall prepare tabular and graphical displays of extracted data and information using templates and tools specified by the Government. The Contractor shall arrange to have access to software for visualizing data (e.g., [Tableau](#)) and drawing chemical structures and/or biological pathways (e.g., ChemSpider, Biocarta, or Ingenuity® Pathway Analysis). The Contractor shall be able to provide outputs from data visualization software in a format specified by the Government (e.g., as a PDF or in an editable format). The Contractor shall arrange for and/or acquire any needed subscriptions or tools upon approval from the CO.

2.2.2 Scoping Reviews

The Contractor shall provide support for developing scoping reviews as needed for a given research question or research concept. For example, brief or targeted scoping reviews may be part of problem formulation to refine the exposures, outcomes or animal models of interest. Scoping reviews may be used to develop a literature search strategy and determine the “scope” or extent of the literature necessary for addressing a research question. Scoping reviews may also be conducted more formally using systematic review-level transparency and documentation to identify and summarize relevant studies on a given topic. Some scoping reviews may be conducted in response to time-sensitive requests, and the Contractor shall have workflows and flexibilities in place to address this need while limiting significant impacts on other project work. Additional details on requirements for literature searching, literature screening, data extraction, and data visualization are available above in Section 2.2.1.

2.2.3 Development and Publication of Systematic Evidence Mapping and Scoping Documents

In coordination with the Government, the Contractor shall develop systematic evidence mapping and scoping documents for publication, including, but not limited to, protocols, peer-reviewed manuscripts, and NIEHS Reports (see Section 2.4). The Contractor shall provide support for compiling documents based on Government guidance, reviewing documents for grammatical correctness, and performing technical review for accuracy and internal consistency to ensure report quality. All documents prepared for publication as Government reports shall be prepared in a predefined template and writing style as described in Section 2.5.

2.2.4 Conflict of Interest Screening

The Contractor shall identify industries and companies involved in the manufacture, distribution, and/or use of specific substances or issues that are the topic of literature-analysis activities or updates for current similar reports. The Government would use this information to identify potentially affected parties for inclusion in conflict-of-interest (COI) screening of employees of the Contractor, subcontractors, consultants, or other interested parties.

2.3 Systematic Review and Evidence Synthesis

The Contractor shall provide staff with the appropriate expertise to manage and support systematic reviews and evidence synthesis for health hazard assessments, including literature

searching, screening, data extraction and visualization, study quality/informativeness assessment, data/evidence interpretation, and scientific evidence integration. Domain-specific expertise for systematic reviews and evidence synthesis shall include toxicology and subdisciplines (e.g., reproductive toxicology, cancer), environmental and occupational epidemiology, human exposure (i.e., occupational hygiene), chemistry (including absorption, distribution, metabolism, and excretion), mechanistic data, and informatics. All systematic review activities to support health hazard assessments shall be conducted according to procedures outlined in the Handbook for Conducting Systematic Reviews for Health Effects Evaluations² and the Handbook for Preparing Report on Carcinogens Monographs³. Some systematic reviews may be conducted in response to time-sensitive requests, and the Contractor shall have workflows and flexibilities in place to address this need while limiting significant impacts on other project work.

2.3.1 Literature Search and Screening

Literature searching and screening to support systematic review and evidence synthesis shall be conducted with the rigor required for systematic review. Additional details on requirements for literature searching and literature screening are available above in Section 2.2.1.

2.3.2 Scientific Data Extraction, Categorization, Management, and Visualization

The Contractor shall extract specified data from relevant scientific studies, as defined by the Government, into databases (e.g., HAWC or Table Builder) using specific web-based forms which are easily exportable to Excel and Word tables. Details on methods for data extraction are provided in the respective handbooks referenced above. Note that these approaches are similar to those outlined in Section 2.2.1, but likely to capture study data in more detail than described above for systematic evidence mapping.

The Contractor shall prepare tabular and graphical displays of extracted data and information using templates, guidance, and tools specified by the Government. The Contractor shall arrange to have access to software for visualizing data (e.g., [Tableau](#)) and drawing chemical structures and/or biological pathways (e.g., ChemSpider, Biocarta, or Ingenuity® Pathway Analysis). The Contractor shall be able to provide outputs from data visualization software in a format specified by the Government (e.g., as a PDF or in an editable format). The Contractor shall arrange for and/or acquire any needed subscriptions or tools upon approval from the CO.

2.3.3 Study Quality/Informativeness Assessment, Data Analysis, Data Interpretation, and Scientific Evidence Integration

The Contractor shall review published scientific studies, summarize relevant information, and assist with assessment of study quality/informativeness including, but not limited to, risk-of-bias assessment. The Contractor shall use scientific knowledge and independent judgment to assist with data analysis (e.g., means or meta-analyses), data interpretation, and evidence integration. The Contractor shall utilize study quality/informativeness assessment methods or frameworks and structured forms specified by the Government to facilitate assessment of study quality/informativeness. The output from the study quality/informativeness assessment shall

² Available at: <https://ntp.niehs.nih.gov/go/ohathandbook>.

³ Available at: <https://ntp.niehs.nih.gov/go/rochandbook>.

be presented in textual, tabular, or graphic formats, with documentation of scientific judgement and the basis of decisions as required.

The Contractor shall also provide statistical and graphing support for data analysis. Examples of this support include, but are not limited to, conducting meta-analyses or preparing stratified forest plots (e.g., using Stata or R) of human epidemiological studies, conducting trend- and pairwise analyses of data from animal studies, and evaluating mechanistic data.

The Contractor shall review the available scientific evidence on a given health effect and use scientific judgment to draft text that critically evaluates and interprets the scientific findings of individual studies and/or integrates the scientific evidence across studies for multiple evidence streams (e.g., human, experimental animal, and mechanistic studies). Evidence evaluation includes assessment of the impact of bias on the study findings.

If requested, the Contractor shall also evaluate and summarize exposure and other related information that may support a health hazard assessment.

2.3.4 Development and Publication of Systematic Review Documents

Research findings from systematic reviews are published in the NTP Monograph report series (noncancer health effects), the Report on Carcinogens (RoC) Monograph report series (cancer health effects), or as peer-reviewed journal publications. As described in Section 2.5, the Contractor shall provide support for compiling documents based on Government guidance, preparing reports in a predefined template and writing style, reviewing documents for grammatical correctness, and performing technical review for accuracy and internal consistency to ensure report quality.

In addition to publication of RoC Monographs, the Contractor shall provide support for publication of the [Report on Carcinogens](#) (RoC). The RoC is a congressionally mandated, science-based, public health report that provides a cumulative list of known or reasonably anticipated human carcinogens; the RoC is updated with new listings every few years. Preparation of each new edition requires updating the information in all or selected profiles, including supporting information, exposure-related information, regulatory information, and other information for all or selected profiles. These activities include conducting information searches, extracting and summarizing relevant information, and integrating the information into the existing profiles. The Contractor shall also provide support for creating individual files for each profile and compiling and publishing the RoC according to Government guidance.

2.3.5 Conflict of Interest Screening

The Contractor shall identify industries and companies involved in the manufacture, distribution, and/or use of specific substances or issues that are the topic of literature-analysis activities or updates for current similar reports. The Government would use this information to identify potentially affected parties for inclusion in COI screening of employees of the Contractor, subcontractors, consultants, or other interested parties.

2.4 Report Development and Scientific Writing

Research findings from DTT, NIEHS, and NTP studies are published in manuscripts and several different report series, including NTP Technical Reports, NTP Toxicity Reports, NTP

Developmental and Reproductive Toxicity (DART) Reports, NTP Immunotoxicity Reports, NTP Research Reports, NIEHS Reports, and other potential report series. Preparation of these documents uses a series of drafts to review and refine data presentation and interpretation, and these reports go through the NIEHS/DTT internal review process continually throughout document preparation. The Contractor shall provide support for statistical analysis of select data, compiling documents based on Government guidance and laboratory study reports, preparing reports in a predefined template and writing style, reviewing documents for grammatical correctness, and performing technical review for accuracy and internal consistency to ensure report quality. In addition, report development and scientific writing support shall be provided for documents post-peer review, including reviewing responses to reviewer comments for accuracy and completeness, revising documents based on peer review comments, and incorporating findings from third-party quality assessment review. Additional details on requirements for document preparation and publication can be found in Section 2.5. Example publications can be found here: <https://ntp.niehs.nih.gov/publications/index.html>.

Support for report development and scientific writing requires access to internal NIEHS databases, as described below in Section 4.

2.5 Document Preparation and Publication

Documents prepared under this contract may be presented as a mix of graphical, tabular, and textual formats. The Contractor shall verify the scientific and technical accuracy of information or data included in all documents whether written by Government staff, the Contractor, or consultants.

The Contractor shall provide administrative and technical support for the preparation and publication of documents, as needed, including writing text using a concise and technically correct format understandable by a wide audience, and assimilating information into reports according to specified style templates and, as required, 508-compliant PDF documents. General administrative and technical skills needed by the Contractor to support this requirement include experience with Microsoft Office, Adobe Acrobat, Section 508-compliance, and records management.

Using a publication style guide provided by the Government, the Contractor shall provide support for formatting documents, copyediting and technical editing all report types, formatting references using software such as EndNote™, and readying reports for publication according to a predefined template format that was previously developed by eXtyles for DTT Government reports. Final documents shall be available in electronic formats and be prepared as 508-compliant PDF documents (see Section 1.6.4). Final 508-compliant documents will be posted to the NTP and/or NIEHS public websites and will also be archived in a shared online location (i.e., SharePoint, Teams, OneDrive). On occasion, the Contractor may be required to post final documents to Government websites.

The Contractor shall provide support for the NIEHS/DTT electronic publishing workflow, including preparation of template and file conversions for web publishing of RoC and NTP monographs, NTP reports, NIEHS reports, and Government documents as identified by the COR. Government-specified software (e.g., eXtyles) shall be utilized to convert final document files to XML for subsequent upload and publication on the National Center for Biotechnology

Information (NCBI) Bookshelf website (<https://www.ncbi.nlm.nih.gov/books/>). The Contractor shall also submit citation information for all final documents to NCBI for indexing within PubMed (<https://pubmed.ncbi.nlm.nih.gov/>).

Occasionally, the Contractor, as required by the COR, shall provide support in obtaining document covers and related graphics. If required, the Contractor shall interface with the NIEHS arts/graphics contractor for their preparation as instructed by the COR. In addition, as required by the COR, the Contractor shall prepare tables and graphs, sometimes utilizing data retrieved from locations such as the [Chemical Effects in Biological Systems](#) database.

In addition to preparing new documents, the Contractor shall provide support for updating existing documents, including reviewing information and content, re-formatting, and preparing documents for electronic publication (e.g., NTP and NIEHS public websites, NCBI Bookshelf/PubMed).

The Contractor shall provide printing services for draft or final documents or products as required and approved by the COR.

2.6 Coordination and Management of External Peer Review

RoC and NTP monographs and NTP and NIEHS reports are published in NCBI's Bookshelf (as described in Section 2.5) and are required to go through external peer review prior to publication. Peer review may occur via individual letter review (like a peer-reviewed manuscript) or via expert panel review at a public meeting. The Contractor shall provide support for external peer review of NIEHS and NTP documents, including identifying experts, developing COI packages, screening potential experts, corresponding with reviewers, preparing review documents, and post-peer review activities. Upon approval from the COR, the Contractor shall obtain and manage subcontracts with identified peer reviewers and pay them for their services.

Experts on a specific topic shall be identified utilizing the online NTP Expert Directory (accessible via the NIH network) as well as publicly available resources. COI packages (including lists of potentially affected companies) shall be prepared based on a given report's topic and all identified experts shall be screened for potential real or perceived COI. The Contractor shall screen and/or assist the Government in obtaining COI information from scientific experts prior to confirmation of service. Procedures developed around expert selection and COI screening should demonstrate how to handle personally identifiable information (PII). In addition, the Contractor shall be familiar with the requirements around influential scientific information (ISI) and highly influential scientific information (HISI) in accordance with the [Office of Management and Budget's Information Quality Bulletin for Peer Review](#) (see Section 7, Applicable References).

2.7 Logistical Management for Advisory Group Meetings and Workshops

Scientific evidence evaluation and knowledge integration activities within NIEHS/DTT may require internal and/or external meetings or events that are remote (web-based, e.g., webcast, Teams, or similar), in-person, or a hybrid of the two, and can vary in size (e.g., 4-6 in-person or up to 200 attendees at a webinar). The types of events can vary and may include advisory group meetings, peer reviews, symposia, workshops, focus/information groups, and seminars. The

events will vary in length, generally from one-half to two days, and will usually be held or coordinated from the NIEHS campus or in the Research Triangle Park, NC or Washington, DC areas, although occasionally they may be held at another domestic or international location. The number of experts required for an event will vary, ranging, for example, from 3-4 speakers for a webinar or symposium up to 10-20 experts for an advisory board meeting. The Government will specify whether the meeting is closed or open to the public and other specific details.

The Contractor shall provide administrative, technical, and logistical support for the planning and execution of these meetings including identification of an appropriate venue, although use of Government space is priority. As requested, the Contractor shall assist the Government in obtaining information from participants for COI screening. The Contractor shall handle travel arrangements for participants, subject to applicable Government regulations. Prior to confirming any arrangements for meetings, the Contractor shall consult with and receive approval from the COR and/or CO. The Contractor shall provide support for managing or preparing materials for the meeting and drafting any needed documents, minutes/reports, notes, or summaries of major points from reports, minutes, or public comments. The Contractor shall also provide support for preparing information for meeting website(s), distributing materials, and handling communications with targeted individuals. Upon approval from the COR, the Contractor shall obtain and manage subcontracts with identified participants and pay them for their services.

Current examples of support for advisory group meetings include support for the NTP Board of Scientific Counselors (BSC), the DTT BSC, the Scientific Advisory Committee on Alternative Toxicological Methods (SACATM), expert panels, and others. The COR will advise the Contractor regarding new iterations of advisory groups, name changes, or other relevant updates, as needed. These advisory groups are federally chartered and provide rigorous scientific peer review and advice on programs and activities. Additional information on these advisory groups can be found here: <https://ntp.niehs.nih.gov/go/advisory>. All advisory meetings are conducted according to the [Federal Advisory Committee Act](#).

2.8 Provision of Scientific or Technical Consultants with Expertise in Relevant Scientific Disciplines

The Contractor shall develop knowledge and key contacts to assist with identifying appropriate experts including consultants, writers, peer reviewers, and speakers. A consultant may be either a Government or non-Government expert who provides information on a specific scientific topic. Areas of scientific expertise for these individuals shall include, but not be limited to, toxicology, epidemiology, pathology, exposure assessment, regulatory processes, and other disciplines. The Contractor shall work with the COR regarding requirements (expertise, COI, timelines, etc.), and make recommendations to the COR regarding appropriate experts. Upon approval from the COR, the Contractor shall obtain and manage subcontracts with identified experts and pay for any services. Logistics for individuals serving as members of peer review panels shall be handled as identified in Sections 2.6 and 2.7.

The Contractor shall maintain and regularly update the online NTP Expert Directory hosted on the NIH network.

2.9 Development and Implementation of Novel Tools and Informatics Approaches

The Contractor shall provide staff with the appropriate expertise to manage and support the development or utilization of novel tools and informatics approaches for scientific evidence evaluation and knowledge integration for translational toxicology. Domain-specific expertise for novel methods shall primarily include data science, informatics, artificial intelligence (AI), and large language models (LLMs), with familiarity and expertise in toxicology and related fields as needed.

As needed, the Contractor shall facilitate the development or utilization of novel approaches for all areas of scientific evidence evaluation and knowledge integration for translational toxicology, with a focus on approaches for literature-based health assessments, evaluating research effectiveness, and vocabularies/ontologies. Support for this activity shall include, but may not be limited to:

- Development and/or implementation of novel tools and informatics approaches
- Evaluation of current approaches, tools, and methodologies
- Adaptation of existing approaches to new workflows
- Support for tools and approaches already being used in existing workflows

The Contractor shall consult with the COR regarding requirements for these activities (expertise, products, timelines, etc.) and make appropriate recommendations regarding how best to address the needed technical assistance. Such activities may be done via consultants to the contract and/or the use of subcontractors. As applicable and directed by the COR, the Contractor shall screen scientific experts for conflict of interest prior to confirmation of service.

2.9.1 Novel Approaches for Literature-based Health Assessments

The Contractor shall provide development and implementation support for tools and approaches applicable to literature-based health assessments, particularly related to improving workflows through the application of AI-enabled approaches (e.g., machine learning, natural language processing) supporting systematic review automation. These activities may include evaluating novel methods for AI-enabled literature screening, evaluating model-based or AI-assisted alternative data extraction techniques, or AI-assisted study quality/informativeness assessment tools.

The Contractor shall provide support for Dextr, an automated data extraction tool with user verification developed for utilization in NIEHS/DTT scoping, systematic evidence mapping, and systematic reviews. Access to Dextr will be provided by the Government. Additional key publications related to Dextr can be found below:

- Walker VR, et al. 2022. [Evaluation of a semi-automated data extraction tool for public health literature-based reviews: Dextr](#). *Environment International*.
- Nowak A, Kunstman P. 2018. [Team EP at TAC 2018: Automating data extraction in systematic reviews of environmental agents](#). Paper presented at: National Institute of Standards and Technology Text Analysis Conference. Gaithersburg, MD.

- Schmitt C, et al. 2018. [Overview of the TAC 2018 Systematic Review Information Extraction Track](#). Paper presented at: National Institute of Standards and Technology Text Analysis Conference. Gaithersburg, MD.

The Contractor shall also provide support for Table Builder, a web-based content management system designed to capture extracted data from the published literature and facilitate qualitative and quantitative analyses (e.g., meta-analyses, statistical testing) for human health assessments. Access to Table Builder will be provided by the Government.

2.9.2 Novel Approaches for Evaluating Research Impact and Effectiveness

The Contractor shall provide development and implementation support for tools and approaches applicable to evaluating the effectiveness and impact of NTP and DTT research (i.e., influencing regulatory actions, citations in other research, and use in scientific and digital platforms). Examples of support may include, but are not limited to, obtaining, evaluating, and documenting bibliometric and visualization tools; preparing and automating literature search strings for NTP and DTT citations; and compiling results into an overall summary of NTP's and DTT's research impacts.

2.9.3 Vocabularies and Ontologies

The Contractor shall provide support for developing and applying approaches for utilization of common, well-defined vocabularies and ontologies in toxicological and environmental health research. Support shall include applying controlled effects vocabularies to extracted data from applicable literature and developing mapping systems among vocabularies, as requested. This work shall be interoperable with the [DTT Data Dictionary](#).

In addition, the Contractor shall provide support for the [Environmental Health Language Collaborative](#) (EHLCC). EHLCC is a NIEHS-supported research initiative to advance community development and application of harmonized language for describing environmental health science research. As needed, the Contractor shall provide program management support including, but not limited to, workshop planning and execution, community engagement, preparation of communication materials, and identification of existing language and data standards, as well as associated gaps.

2.10 Program Management

Coordination of DTT's research portfolio requires program management support for both Divisional activities and interagency activities supporting NTP.

2.10.1 Support for Scientific Program Management

The Contractor shall provide general program management support for the scientific activities of DTT's research programs. Examples of relevant support include, but are not limited to, communication and coordination of activities and deadlines, support for meetings, information and knowledge management for given scientific and organizational topics, metric analytics, summarizing milestones and outputs, and preparing and/or updating the scientific content of web- and non-web-based communication materials.

2.10.2 Support for the National Toxicology Program

Programmatic and policy oversight of NTP activities and research programs is currently coordinated through the Office of NTP Scientific Operations and Coordination (ONSOC), which includes senior federal staff from each respective agency. The COR will advise the Contractor of relevant organizational changes, as needed. The Contractor shall provide program and project management support for ONSOC activities. Examples of relevant support include, but are not limited to, communication and coordination of activities and deadlines, support for interagency meetings, and programmatic support for NTP work groups on given scientific topics.

3 Operating Procedures

3.1 Project Management

The Contractor shall provide project management for the contract to include, but not be limited to:

- Transitioning into and out of the contract with minimal disruption
- Invoicing accurately and on time
- Submitting deliverables and reports and otherwise complying with contractual commitments
- Providing transparency into operations for carrying out tasks
- Managing personnel and staffing
- Coordinating and scheduling human and other resources
- Providing project management and product tracking
- Providing timely communication with the COR

The Contractor shall have in place an organizational structure, policies, procedures, processes, and resources to plan and coordinate preparation of the work carried out under this contract. The Contractor shall provide the COR a plan for maintaining the accuracy and integrity of data, information, and work products at all levels in a project including version control of documents, extraction of information, and management and organization of project files. The Contractor shall establish internal procedures to evaluate the quality of the work and shall take actions and communicate to the Government plans to ensure the production of high-quality products. This plan shall also include management strategies to maintain talented staff through the period of performance of this contract. This plan shall be approved by the COR. Work performed across multiple functional groups shall be coordinated in a way that provides independence for peer reviews. Any incident or unforeseeable occurrence not listed in this document, or any change in personnel that might have an impact on fulfilling the requirements as described in the SOW, shall be reported immediately to the COR.

The PI and PM shall be responsible for the daily management of the contract and serve as points of contact for communication with the Government officials. Collectively, the PI and PM are expected to be knowledgeable and involved in all aspects of the contract, including project status, record keeping, reporting, and detailed analysis of budget for all contract functions.

Although the contract as a whole shall be for the ongoing support of scientific evidence evaluation and knowledge integration for translational toxicology, a research outline will be

prepared and issued by the Government for each activity to be performed under the contract. Each research outline will include a detailed description of the work to be performed or an outline of project objectives. In response, the Contractor shall submit a technical work plan outlining the proposed technical approach and a budget proposal describing the proposed budget and level of effort. Information to be included in the technical work plan and budget are included in Technical Exhibit A of this SOW. Work on a given activity shall begin upon receipt of the research outline from the Government and a work plan shall be submitted to the Government following research outline receipt. During execution of an activity, if the need for support of a particular research outline changes, a modified research outline shall be submitted by the Government to the Contractor; if the revisions are significant the Government may request technical and/or budget proposals for the revision.

Administrative activities shall be conducted by the Contractor as described in an administrative research outline. These activities include general management, periodic reports, or other Government-approved work not directly attributable to a specific research outline.

The Contractor shall use a DTT program and project management (PPM) software (i.e., Microsoft Dynamics 365) specified by the Government to document and update research activities, including, but not limited to, research outlines submitted by the Government, technical work plans and budget information, milestone schedules, work status, interim data (in the form of tables, graphs, spreadsheets, and/or other documents), and deliverables. All substantive communication and decisions throughout the performance of work on an activity shall be documented within the software.

Over the course of a given activity, the Government anticipates that the processes used to develop documents and research products and carry out related activities may change as program-related changes are implemented. Additionally, updates to the instructions for a specific activity may be made by the Government due to changing requirements or in particular cases due to the unique or complex nature of a particular activity. Any updates to processes or instructions will be communicated by the Government to the Contractor.

3.2 Data Management

The Contractor shall provide a data management plan for the management, sharing, backup, storage, and security of all paper and electronic data, files, and information collected and generated as part of completing the products outlined in this SOW. This plan includes the separate storage and efficient retrieval of data from individual products. This data management plan shall adhere to the U.S. Department of Health and Human Services (DHHS) Office of the Chief Information Officer (OCIO) policy⁴ and be approved by the COR. The Contractor shall keep all data and documents used or prepared confidential until Government publication. Once projects are completed, all paper and electronic files shall be transferred to the Government upon the COR's request for long term storage.

⁴ U.S. Department of Health and Human Services Office of the Chief Information Officer Policy: <https://www.hhs.gov/about/agencies/asa/ocio/cybersecurity/index.html>.

All data derived or collected by Contractor shall reside in a data management system approved by the Government after acceptance by the COR. The Contractor shall use IDs and metadata interoperable with those used in the NIEHS/DTT data ecosystem for interoperability, accuracy, and reliability across the NIEHS/DTT data ecosystem. Metadata shall meet DTT standards for discoverability and re-use. Sensitive data shall be stored securely in accordance with patient agreements. All research data products shall be registered in the DTT Research Data Catalog with metadata to facilitate discoverability. Code shall be stored in government repositories such as, but not limited to, GitHub as directed by the COR.

The Government will have unlimited data rights ownership of all data and information generated during the Contract. The Government retains the right to access all information and data generated under this Contract at any time.

3.3 Subcontractors and Consultants

The Contractor, as directed by the COR, shall identify, and propose qualified organizations to conduct the work as subcontractor(s) and present to the CO and the COR for approval. A detailed written subcontract plan shall be provided to the CO and COR. The subcontract plan shall include the area of expertise and related peer-reviewed journal articles authored by the proposed consultant relevant to the request, as well as cost and a detailed timeline of accomplishing the project. Only upon approval by the CO shall work at the subcontract organization begin. The Contractor shall ensure that no institutional or individual COI issues exist for the coordination and/or selection of the organizations participating in scientific evidence evaluation and knowledge integration activities.

The Contractor shall summarize subcontractor performance and include the summary in the Contractor's monthly progress report to the COR. Information included in this summary shall consist of work performed by the subcontractor including deliverables, level of effort, and costs expended during the past month, etc. In addition, the Contractor shall provide a description of how the subcontract aided the Contractor in completing or nearing completion of a specific project within the monthly progress report. Once the specified project is completed, no further work shall be given to the subcontractor until a new detailed project plan is presented to the COR.

The Contractor shall ensure that the employees of the prime contract and any subcontractors work together as a unified team. It should be transparent to the Government whether a contract employee works for the prime or a subcontractor. The Contractor and any subcontractors shall conduct or provide to their employees detailed instruction on government statutes, regulations, policies, and guidelines in areas such as employee conduct, ethics, safety, security, health, fire prevention, and the environment as they pertain to this contract.

3.4 Contractor Travel

The Contractor's PI, PM, and/or Contractor staff shall attend meetings (primarily via remote options) with the COR and/or other Government staff as determined by the Contractor's PI and the COR. In-person meetings (e.g., with Government staff or external meetings such as workshops and peer reviews) shall typically be held at NIEHS or in the Research Triangle Park, NC or Washington, DC areas, although occasionally these meetings may be held at another domestic or international locations. In addition, the PI and Contractor staff shall be required to

consult on an as needed basis with the COR and other COR-identified Government staff, either in-person or via remote meeting options. These meetings may be announced on short notice. Contractor staff may also be required to conduct site visits of event venues (at NIEHS, or in the Research Triangle Park, NC, Washington, DC, or designated areas). Travel associated with the contract shall be reimbursed in accordance with federal travel regulations; all travel requests must be submitted to the COR and CO for approval prior to traveling.

3.5 Contract Management and Monitoring

Contract performance will be monitored through meetings with the PI, PM, and senior staff, review of monthly reports and research deliverables, electronic mail, and telephone discussion, as required.

The Contractor and Government staff shall have formal meetings, typically biweekly, although more frequent meetings to discuss specific areas of a given activity shall occur on an as needed basis, and possibly as frequent as once or twice a week. The COR, on a case-by-case basis, shall notify the Contractor if more frequent meetings are needed. In addition, the Contractor shall communicate with Government staff by phone or via email. Group meetings with the Contractor may take place in-person at NIEHS facilities or via remote meeting options (i.e., Zoom, Teams). Due to the broad nature of the activities supported by the contract requirements, Government project leads will consist of staff from different branches and offices within DTT. Additional information about the DTT organizational structure is available on the [NIEHS website](#).

3.6 Contractor Responsibilities

The Contractor shall arrange for or acquire any subscriptions for information resources, information management, project management, or electronic publication software (e.g., for Web publishing) following discussion and approval by the COR and/or CO. The Contractor shall arrange for any associated arts/graphics or printing. All software and databases used to carry out the project objectives and requirements in this contract shall be specified and approved by the CO and the COR. All software and databases used or secured must be PC and MAC compatible unless otherwise specified. All electronic deliverables must be PC and MAC compatible unless otherwise specified. The Contractor shall ensure section 508-compliance of all final electronic deliverables according to Adobe Acrobat version 11 [see §1194.22 Web-based Intranet and Internet Information and Applications (<http://www.section508.gov/>)] and Section 508 of the Rehabilitation Act of 1973, as amended, 29 U.S.C. 794d) or a current version identified by NIEHS as the standard.

The Contractor shall ensure that all activities are carried out and that all deliverables adhere to COR-approved instructions or specifications for content and format. All deliverables shall follow NTP or NIEHS website specifications where applicable; be technically accurate and scientifically sound; provide transparent, systematic, and clear interpretation of the scientific information; address the relevant scientific issues; be free of grammatical and spelling errors and be reviewed for readability and visual appeal. The Contractor shall ensure section 508-compliance of all final electronic deliverables as noted above.

Before beginning a project for a specific substance or issue, the Contractor shall complete the COI requirement. The COR will provide the Contractor instructions on the project.

4 Government Furnished Property and Services

4.1 Supplies and Services

Upon proper security clearance (e.g., [Homeland Security Presidential Directive 12](#) [HSPD-12]), the Government shall provide the Contractor access to the NIEHS and NIH electronic resources including, but not limited to, Citrix and Virtual Private Network (VPN) client software to access the NIH and NIEHS networks, [NIEHS and NIH library resources](#), content management software for interacting with NIEHS-hosted websites, and relevant NIEHS/DTT databases.

If required to perform work under the Contract, Contractor staff shall obtain NIEHS security badges and an NIH account. All Contractor staff with NIEHS security badges shall adhere to DHHS OCIO⁵ and NIH/NIEHS requirements for connecting to the NIEHS IT resources. Contract staff shall also complete all annual refresher training associated with the use of an NIH account in a timely manner.

4.2 Facilities

All work shall be conducted at the Contractor's facility. Some work will require attending meetings at NIEHS or in the Washington, DC area. The NIEHS main campus is located at 111 T.W. Alexander Drive, Research Triangle Park, NC 27709 and the NIEHS satellite campus is currently located at 530 Davis Drive, Morrisville, NC 27650.

4.3 Software/Training

The Government will provide licensed access to a Government-specified program and project management software (currently Microsoft Dynamics 365). DTT will provide contract staff with the necessary training prior to accessing the system.

4.4 Equipment

If required to perform activities under the Contract, the Government will provide a secure laptop computer in order for the Contractor to have access to the VPN in use at NIEHS, given the proper security clearances are processed for the Contractor staff using the equipment. Contractor staff who are granted access to the NIEHS VPN will be required to comply with all NIH training requirements for the maintenance of these accounts.

5 Contractor Furnished Property and Services

5.1 General

The Contractor shall provide adequate office space, equipment, and services to perform the activities described in this SOW.

⁵ U.S. Department of Health and Human Services Office of the Chief Information Officer Policy: <https://www.hhs.gov/about/agencies/asa/ocio/cybersecurity/index.html>.

5.2 Equipment and Software

The Contractor shall provide all equipment necessary to fulfill the requirements of the SOW that are not explicitly specified as being provided by the Government. These materials include but are not limited to computers, printers, phones, Microsoft Office software, web-based conferencing software, and Adobe Acrobat Professional (for Section 508 compliance).

Equipment used for access, such as computers, printers, and software, shall support Internet and Intranet connections, be compatible with NIH computer operating systems, and comply with NIH security policies. The Contractor shall have appropriate and reliable internet connectivity to complete work. The Contractor shall conform to security requirements or usability guidelines as appropriate.

The Contractor shall maintain compatible versions of all software applications and licenses and shall provide personnel properly trained with the latest releases and upgrades.

6 Definitions and Acronyms

6.1 Definitions

Although not inclusive of every term used within this SOW, the following provides a list of definitions used throughout this SOW and commonly used in the acquisition field.

Contracting Officer (CO) – means a person with authority to enter into, administer, and or terminate contracts, and make related determinations and findings on behalf of the Government. Note: The only individual who can legally bind the Government.

Contracting Officer's Representative (COR) – as defined in DFARS 202.101, means an individual designated and authorized in writing by the Contracting Officer to perform specific technical or administrative functions. Any individual delegated responsibilities pursuant to subpart FAR 1.602-2 of Reference (e), regardless of local terminology, must be certified in accordance with this instruction. Local terminology may include COR or Alternate COR. These other surveillance support personnel may serve as on-site representatives of the COR in performance of actual contract surveillance if they meet all COR requirements and have been appointed by the Contracting Officer as alternate CORs.

Contractor – means a supplier or vendor awarded a contract to provide specific supplies or service to the Government. The term used in this contract refers to the prime.

Contractor-acquired Property – means property acquired, fabricated, or otherwise provided by the Contractor for performing a contract, and to which the Government has title.

Day – means, unless otherwise specified, a calendar day.

Defective Service – means service output that does not meet the standard of performance associated with the SOW.

Deliverable – means anything that can be physically delivered but may include non-manufactured things such as meeting minutes or reports.

Government-furnished Property – means property furnished to the Contractor by the Government and subsequently furnished to the Contractor for performance of a contract.

Government-furnished property includes, but is not limited to, spares and property furnished for repair, maintenance, overhaul, or modification. Government-furnished property also includes Contractor-acquired property if the Contractor-acquired property is a deliverable under a cost contract when accepted by the Government for continued use under the contract.

Government Property – means all property owned or leased by the Government. Government property includes both Government-furnished property and Contractor-acquired property. Government property includes material, equipment, special tooling, special test equipment, and real property. Government property does not include intellectual property and software.

Physical Security – means that part of security concerned with physical measures designed to safeguard personnel; to prevent unauthorized access to equipment, installations, material, and documents; and to safeguard against espionage, sabotage, damage, and theft.

Quality Control – means all necessary measures taken by the Contractor to assure that the quality of an end product or service shall meet contract requirements.

Subcontractor – means one that enters into a contract with a prime Contractor. The Government does not have privity of contract with the subcontractor.

Work Day – The number of hours per day the Contractor provides services in accordance with the contract.

Work Week – Monday through Friday, unless otherwise specified.

6.2 Acronyms

Although not inclusive of every term used within this SOW, the following provides a list of acronyms used throughout this SOW or commonly used in the acquisition field.

BSC	Board of Scientific Counselors
CFR	Code of Federal Regulations
CO	Contracting Officer
COI	conflict-of-interest
COR	Contracting Officer's Representative
CS	Contract Specialist
DART	developmental and reproductive toxicity
DTT	Division of Translational Toxicology
EHLC	Environmental Health Language Collaborative
EIT	electronic and information technologies
FACA	Federal Advisory Committee Act
FAR	Federal Acquisition Regulation
HAWC	Health Assessment Collaborative Workspace
HISI	highly influential scientific information
ICT	information and communication technology
ISI	influential scientific information
IT	information technology
PI	Principal Investigator
PM	Program Manager

NCBI	National Center for Biotechnology Information
NCTR	National Center for Toxicological Research
NIEHS	National Institute of Environmental Health Sciences
NIH	National Institutes of Health
NIOSH	National Institute for Occupational Safety and Health
NTP	National Toxicology Program
ONSOC	Office of NTP Scientific Operations and Coordination
PII	personally identifiable information
RoC	Report on Carcinogens
SACATM	Scientific Advisory Committee on Alternative Toxicological Methods
SOW	statement of work
USC	United States Code
VPN	Virtual Private Network

7 Applicable References

The following publications, manuals, regulations, etc. are mentioned in this SOW and are listed below:

- Handbook for Conducting Systematic Reviews for Health Effects Evaluations
 - Available at: <https://ntp.niehs.nih.gov/go/ohathandbook>
- Handbook for Preparing Report on Carcinogens Monographs
 - Available at: <https://ntp.niehs.nih.gov/go/rochandbook>
- Influential and Highly Influential Scientific Information
 - Available on the [Office of Management and Budget's Information Quality Bulletin for Peer Review](#)
- Federal Advisory Committee Act
 - Available at: <https://www.doi.gov/execsec/faca>
- Homeland Security Presidential Directive 12
 - Available at: <https://www.dhs.gov/homeland-security-presidential-directive-12>
- U.S. DHHS OCIO Policy
 - Available at: <https://www.hhs.gov/about/agencies/asa/ocio/cybersecurity/index.html>

8 Technical Exhibits

8.1 Technical Exhibit A. Technical Work Plan and Budget

A. Technical work plans shall include (at a minimum):

- Detailed staffing plan, including program manager, research outline manager, professional staff, technical staff, and applicable subject matter experts. If the technical work plan includes staff that have not previously performed work on the contract, the work plan shall be accompanied by the requisite documentation of personnel experience and training.
- Detailed description of how all research outline requirements will be accomplished.

- Schedule for major milestones and deliverables throughout the life of an activity.

B. Budget proposals shall include (at a minimum):

- Significant technical assumptions that may impact effort or cost.
- Projected costs of purchased materials, other direct costs, etc.
- Detailed hours broken down by individually named professional staff and technical staff by category.
- Estimated level of effort must include a detailed breakdown of cost (direct, indirect, and fee) and hours (by labor category) by month for each category and must align with the milestone schedule in the technical work plan and contract period.
- Hours and dollars for any subcontracted work or purchased service shall be accompanied by the subcontractor quote.

APPENDIX A – HHS SECURITY AND PRIVACY LANGUAGE FOR INFORMATION AND IT PROCUREMENTS

SECTION A – INFORMATION SECURITY AND/OR PHYSICAL ACCESS SECURITY

1. Baseline Security Requirements

- a. **Applicability.** The requirements herein apply whether the entire contract or order (hereafter "contract"), or portion thereof, includes either or both of the following:
 - i. **Access (Physical or Logical) to Government Information:** A Contractor (and/or any subcontractor) will have or will be given the ability to have, routine physical (entry) or logical (electronic) access to government information.
 - ii. **Operate a Federal System Containing Information:** A Contractor (and/or any subcontractor) will operate a federal system and information technology containing data that supports the HHS mission. In addition to the Federal Acquisition Regulation (FAR) Subpart 2.1 definition of "information technology" (IT), the term as used in this section includes computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the central processing unit of a computer, software, firmware and similar procedures, services (including support services), and related resources.
- b. **Safeguarding Information and Information Systems.** All government information and information systems must be protected in accordance with HHS/NIH policies and level of risk. At a minimum, the Contractor (and/or any subcontractor) must:
 - i. Protect the:
 - **Confidentiality**, which means preserving authorized restrictions on access and disclosure, based on the security terms found in this contract, including means for protecting personal privacy and proprietary information;
 - **Integrity**, which means guarding against improper information modification or destruction, and ensuring information non-repudiation and authenticity; and
 - **Availability**, which means ensuring timely and reliable access to and use of information.
 - ii. Categorize all information owned and/or collected/managed on behalf of HHS/NIH and information systems that store, process, and/or transmit HHS information in accordance with FIPS 199 and National Institute of Standards and Technology ([NIST Special Publication \(SP\) 800-60, Volume II: Appendices to Guide for Mapping Types of Information and Information Systems to Security Categories](#)). Based on information provided by the NIEHS ISSO, NIEHS SOP, or other representative, the impact level for each Security Objective (Confidentiality, Integrity, and Availability)

and the Overall Impact Level, which is the highest watermark of the three factors of the information or information system are the following:

- § **Confidentiality:** ☒ Low ☐ Moderate ☐ High
- § **Integrity:** ☒ Low ☐ Moderate ☐ High
- § **Availability:** ☒ Low ☐ Moderate ☐ High
- § **Overall Impact Level:** ☒ Low ☐ Moderate ☐ High

- iii. Based on the agreed-upon level of impact, implement the necessary safeguards to protect all information systems and information collected and/or managed on behalf of HHS/NIH regardless of location or purpose.
 - iv. Report any discovered or unanticipated threats or hazards by either the agency or contractor, or if existing safeguards have ceased to function immediately after discovery, **within one (1) hour or less**, to the government representative(s).
 - v. Adopt and implement all applicable policies, procedures, controls, and standards required by the HHS/NIH Information Security Program to ensure the confidentiality, integrity, and availability of government information and government information systems for which the Contractor is responsible under this contract or to which the Contractor may otherwise have access under this contract. Obtain all applicable security and privacy policies by contacting the CO/COR or HHS/NIH security and/or privacy officials.
- c. **Privacy Compliance.** Comply with the E-Government Act of 2002, NIST SP 800-53, and applicable HHS/NIH/NIEHS privacy policies, and complete all the requirements below:
- i. Per the Office of Management and Budget (OMB) Circular A-130, Personally Identifiable Information (PII), is "information that can be used to distinguish or trace an individual's identity, either alone or when combined with other information that is linked or linkable to a specific individual." Examples of PII include, but are not limited to the following: Social Security number, date and place of birth, mother's maiden name, biometric records, etc.
 - ii. Based on information provided by the ISSO, system/data owner, or other security or privacy representative, it has been determined that this solicitation/contract involves:

☐ No PII ☒ PII
 - iii. The Contractor must support the agency with conducting a Privacy Threshold Analysis (PTA) for the information system and/or information handled under this contract to determine whether or not a full Privacy Impact Assessment (PIA) needs to be completed.
 - § If the results of the PTA show that a full PIA is needed, the Contractor must support the agency with completing a PIA for the system or information within **60 days** after completion of the PTA and in accordance with HHS policy and OMB M-03-22, *Guidance for Implementing the Privacy Provisions of the E-Government Act of 2002*.

- § The Contractor must support the agency in reviewing the PIA at least every **three years** throughout the system development lifecycle (SDLC)/information lifecycle, or when determined by the agency that a review is required based on a major change to the system, or when new types of PII are collected that introduces new or increased privacy risks, whichever comes first.
- d. **Controlled Unclassified Information (CUI). Executive Order 13556 defines CUI** as "information that laws, regulations, or Government-wide policies require to have safeguarding or dissemination controls, excluding classified information." The Contractor (and/or any subcontractor) must comply with *Executive Order 13556, Controlled Unclassified Information, (implemented at 3 CFR, part 2002)* when handling CUI. 32 C.F.R. 2002.4(aa) As implemented the term "*handling*" refers to "...any use of CUI, including but not limited to marking, safeguarding, transporting, disseminating, re-using, and disposing of the information." 81 Fed. Reg. 63323. The requirements below apply only to nonfederal systems that process, store, or transmit CUI, or that provide security protection for such components. All sensitive information that has been identified as CUI by a regulation or statute, handled by this solicitation/contract, must be:
- i. Marked appropriately;
 - ii. Disclosed to authorized personnel on a Need-To-Know basis;
 - iii. Protected in accordance with NIST SP 800-53, *Security and Privacy Controls for Information Systems and Organizations* applicable baseline if handled by a Contractor system operated on behalf of the agency, or NIST SP 800-171, *Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations* if handled by internal Contractor system; and
 - iv. Returned to HHS control, destroyed when no longer needed, or held until otherwise directed. Information and/or data must be disposed of in accordance with NIST SP 800-88, *Guidelines for Media Sanitization*.
- e. **Protection of Sensitive Information.** For security purposes, information is *or* may be sensitive because it requires security to protect its confidentiality, integrity, and/or availability. The Contractor (and/or any subcontractor) must protect all government information that is *or* may be sensitive by securing it with a solution that is validated with current FIPS 140 validation certificate from the NIST CMVP.
- f. **Confidentiality and Nondisclosure of Information.** Any information provided to the contractor (and/or any subcontractor) by HHS or collected by the contractor on behalf of HHS must be used only for the purpose of carrying out the provisions of this contract and must not be disclosed or made known in any manner to any persons except as may be necessary in the performance of the contract. The Contractor assumes responsibility for protection of the confidentiality of Government records and must ensure that all work performed by its employees and subcontractors must be under the supervision of the Contractor. Each Contractor employee or any of its subcontractors to whom any HHS records may be made available or disclosed must be notified in writing by the

Contractor that information disclosed to such employee or subcontractor can be used only for that purpose and to the extent authorized herein.

The confidentiality, integrity, and availability of such information must be protected in accordance with HHS and NIH policies. Unauthorized disclosure of information will be subject to the HHS/NIH sanction policies and/or governed by the following laws and regulations:

- i. 18 U.S.C. 641 (Criminal Code: Public Money, Property or Records);
 - ii. 18 U.S.C. 1905 (Criminal Code: Disclosure of Confidential Information); and
 - iii. 44 U.S.C. Chapter 35, Subchapter I (Paperwork Reduction Act).
- g. **Internet Protocol Version 6 (IPv6).** All procurements using Internet Protocol must comply with OMB Memorandum M-05-22, *Transition Planning for Internet Protocol Version 6 (IPv6)*.
- h. **Information and Communications Technology (ICT).** ICT products and services from prohibited entities/sources must not be used/acquired in compliance with Public Law 115-232, Section 889 Parts A and B, FAR 4.21, FAR 52.204.23, FAR 52.204.24, and FAR 52.204.25. The contractor (and/or any subcontractor) must notify the government if they identify prohibited ICT products and/or services are used during the contract performance.
- i. **Government Websites.** All new and existing public-facing government websites must be securely configured with Hypertext Transfer Protocol Secure (HTTPS) using the most recent version of Transport Layer Security (TLS). In addition, HTTPS must enable HTTP Strict Transport Security (HSTS) to instruct compliant browsers to assume HTTPS at all times to reduce the number of insecure redirects and protect against attacks that attempt to downgrade connections to plain HTTP. For internal-facing websites, HTTPS is not required, but it is highly recommended. Consult the *HHS Policy for Internet and Email Security* for additional information.
- j. **Contract Documentation.** The Contractor must use provided templates, policies, forms and other agency documents. NIH will specify which documents/forms will be provided to comply with contract deliverables as appropriate.
- k. **Standard for Encryption.** The Contractor (and/or any subcontractor) must:
- i. Comply with the *HHS Standard for Encryption of Computing Devices and Information* to prevent unauthorized access to government information.
 - ii. Encrypt all sensitive federal data and information (i.e., PII, protected health information [PHI], proprietary information, etc.) in transit (i.e., email, network connections, etc.) and at rest (i.e., servers, storage devices, mobile devices, backup media, etc.) with encryption solution that is validated with current FIPS 140 validation certificate from the NIST CMVP.
 - iii. Secure all devices (i.e.: desktops, laptops, mobile devices, etc.) that store and process government information and ensure devices meet HHS and NIH-specific

encryption standard requirements. Maintain a complete and current inventory of all laptop computers, desktop computers, and other mobile devices and portable media that store or process sensitive government information (including PII).

- iv. Verify that the encryption solutions in use have been validated under the Cryptographic Module Validation Program to confirm compliance with current FIPS 140 validation certificate from the NIST CMVP. The Contractor must provide a written copy of the validation documentation to the COR within **15 days** of the validation.
 - v. Use the Key Management system on the HHS personal identification verification (PIV) card or establish and use a key recovery mechanism to ensure the ability for authorized personnel to encrypt/decrypt information and recover encryption keys <http://csrc.nist.gov/publications/>. Encryption keys must be provided to the COR upon request and at the conclusion of the contract.
 - l. **Contractor Non-Disclosure Agreement (NDA).** Each Contractor (and/or any subcontractor) employee having access to non-public government information under this contract must complete the NIH/NIEHS non-disclosure agreement, <https://ocio.nih.gov/aboutus/publicinfosecurity/acquisition/Documents/Nondisclosure.pdf>, as applicable. Contractors (and/or subcontractors) must submit a copy of each signed and witnessed NDA to the Contracting Officer (CO) and/or CO Representative (COR) prior to performing any work under this acquisition.
2. Training Requirements
- a. **Mandatory Training for All Contractor Staff.** All Contractor (and/or any subcontractor) employees assigned to work on this contract must complete the applicable HHS/NIH/NIEHS Contractor Information Security Awareness, Privacy, and Records Management training (provided upon contract award) before performing any work under this contract. Thereafter, the employees must complete <https://irtsectraining.nih.gov/> Information Security Awareness, Privacy, and Records Management training at least **annually**, during the life of this contract. All provided training must be compliant with HHS training policies.
 - b. **Role-based Training.** All Contractor (and/or any subcontractor) employees with significant security responsibilities (as determined by the program manager) must complete role-based training **annually** commensurate with their role and responsibilities in accordance with HHS policy and the *HHS Role-Based Training (RBT) of Personnel with Significant Security Responsibilities Memorandum*. Read further guidance about the NIH Role-based Training at: <https://www.hhs.gov/about/agencies/asa/ocio/cybersecurity/security-awareness-training/index.html>.
 - c. **Training Records.** The Contractor (and/or any subcontractor) must maintain training records for all its employees working under this contract in accordance with HHS policy.

A copy of the training records must be provided to the CO and/or COR within **30 days** after contract award and **annually** thereafter or upon request.

3. Rules of Behavior

- a. The Contractor (and/or any subcontractor) must ensure that all employees performing on the contract comply with the *HHS Information Technology General Rules of Behavior*, *HHS Rules of Behavior for Privileged Users*.
- b. All Contractor employees performing on the contract must read and adhere to the Rules of Behavior before accessing Department data or other information, systems, and/or networks that store/process government information, initially at the beginning of the contract and at least **annually** thereafter, which may be done as part of annual NIH Information Security Awareness Training. If the training is provided by the contractor, the signed ROB must be provided as a separate deliverable to the CO and/or COR per defined timelines above.

4. Incident Response

- a. The Contractor (and/or any subcontractor) must respond to all alerts/Indicators of Compromise (IOCs) provided by HHS Computer Security Incident Response Center (CSIRC)/NIH IRT teams **within 24 hours**, whether the response is positive or negative.

FISMA defines an incident as "an occurrence that (1) actually or imminently jeopardizes, without lawful authority, the integrity, confidentiality, or availability of information or an information system; or (2) constitutes a violation or imminent threat of violation of law, security policies, security procedures, or acceptable use policies. In accordance with OMB M-17-12, *Preparing for and Responding to a Breach of Personally Identifiable Information (PII)*, an incident is "an occurrence that (1) actually or imminently jeopardizes, without lawful authority, the integrity, confidentiality, or availability of information or an information system; or (2) constitutes a violation or imminent threat of violation of law, security policies, security procedures, or acceptable use policies" and a privacy breach is "the loss of control, compromise, unauthorized disclosure, unauthorized acquisition, or any similar occurrence where (1) a person other than an authorized user accesses or potentially accesses personally identifiable information or (2) an authorized user accesses or potentially accesses personally identifiable information for an other than authorized purpose." For additional information on the HHS breach response process, please see the *HHS Policy and Plan for Preparing for and Responding to a Breach of Personally Identifiable Information (PII)*."

- b. In the event of a suspected or confirmed incident or breach, the Contractor (and/or any subcontractor) must:
 - i. Protect all sensitive information, including any PII created, stored, or transmitted in the performance of this contract, with encryption solution that is validated with current FIPS 140 validation certificate from the NIST CMVP.
 - ii. NOT notify affected individuals unless so instructed by the Contracting Officer or designated representative. If so instructed by the Contracting Officer or

representative, the Contractor must send NIH approved notifications to affected individuals in accordance with [https://wiki.ocio.nih.gov/wiki/index.php/US-CERT Federal Incident Notification Guidelines](https://wiki.ocio.nih.gov/wiki/index.php/US-CERT_Federal_Incident_Notification_Guidelines).

- iii. Report all suspected and confirmed information security and privacy incidents and breaches to the NIEHS ISSO team via e-mail at isso@niehs.nih.gov, NIH Incident Response Team (IRT) via email at IRT@mail.nih.gov, COR, CO, NIH/NIEHS SOP (or his or her designee), and other stakeholders, including breaches involving PII, in any medium or form, including paper, oral, or electronic, as soon as possible and without unreasonable delay, no later than **one (1) hour**, and consistent with the applicable NIH/NIEHS and HHS policy and procedures, NIST standards and guidelines, as well as US-CERT notification guidelines. The types of information required in an incident report must include at a minimum: company and point of contact information, contact information, impact classifications/threat vector, and the type of information compromised. In addition, the Contractor must:
 - Cooperate and exchange any information, as determined by the Agency, necessary to effectively manage or mitigate a suspected or confirmed breach;
 - Not include any sensitive information in the subject or body of any reporting e-mail; and
 - Encrypt sensitive information in attachments to email, media, etc.
- iv. Comply with OMB M-17-12, *Preparing for and Responding to a Breach of Personally Identifiable Information*, and HHS/NIH and NIH privacy breach response policies when handling PII breaches.
- v. Provide full access and cooperate on all activities as determined by the Government to ensure an effective incident response, including providing all requested images, log files, and event information to facilitate rapid resolution of sensitive information incidents. This may involve disconnecting the system processing, storing, or transmitting the sensitive information from the Internet or other networks or applying additional security controls. This may also involve physical access to contractor facilities during a breach/incident investigation within an **hour** of discovery.

5. Position Sensitivity Designations

All Contractor (and/or any subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR). The following position sensitivity designation levels apply to this solicitation/contract:

- ☐ Tier 5: Critical Sensitive and Special Sensitive National Security, including Top Secret, SCI, and "Q" access eligibility.
- ☐ Tier 5SR: Reinvestigation.
- ☐ Tier 4: High Risk Public Trust (HRPT).

- ☐ Tier 4SR: Reinvestigation.
- ☐ Tier 3: Non-Critical Sensitive, National Security, including Secret and "L" access eligibility.
- ☐ Tier 3SR: Reinvestigation.
- ☒ Tier 2S with Subject Interview: Moderate Risk Public Trust (MRPT).
- ☐ Tier 2SR: Reinvestigation.
- ☐ Tier 1: Low Risk, Non-Sensitive, including HSPD-12 Credentialing.

6. Homeland Security Presidential Directive (HSPD)-12

The Contractor (and/or any subcontractor) and its employees must comply with Homeland Security Presidential Directive (HSPD)-12, *Policy for a Common Identification Standard for Federal Employees and Contractors*; OMB M-05-24; OMB M-19-17; FIPS 201, *Personal Identity Verification (PIV) of Federal Employees and Contractors*; HHS HSPD-12 policy; and *Executive Order 13467, Part 1 §1.2*.

For additional information, see HSPD-12 policy at: <https://www.dhs.gov/homeland-security-presidential-directive-12>.

7. Roster

The Contractor (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, have the ability to access, or host and/or maintain a government information system(s). The roster must be submitted to the COR and/or CO within **fourteen (14) calendar days** of the effective date of this contract. Any revisions to the roster as a result of staffing changes must be submitted within **seven (7) calendar days** of the change. The COR will notify the Contractor of the appropriate level of investigation required for each staff member. An electronic template, "Roster of Employees Requiring Suitability Investigations," is available for Contractor use at: - <https://oamp.od.nih.gov/nih-document-generation-system/dgs-workform-information/attachment-files-section-j>

If the employee is filling a new position, the Contractor must provide a position description and the Government will determine the appropriate suitability level. Upon receipt of the Government's notification of applicable Suitability Investigations required, the Contractor shall complete and submit the required forms within 30 days of the notification.

8. Contract Initiation and Expiration

- a. **General Security Requirements.** The Contractor (and/or any subcontractor) must comply with information security and privacy requirements, Enterprise Performance Life Cycle (EPLC) processes, HHS Enterprise Architecture requirements to ensure information is appropriately protected from initiation to expiration of the contract. All information systems development or enhancement tasks supported by the contractor must follow the HHS EPLC framework and methodology and in accordance with the HHS Contract Closeout Directive (2018) located at: <https://oamp.od.nih.gov/sites/default/files/DGS/contracting-forms/HHS-Closeout-Directive-2018.pdf>. HHS EA requirements are located at:

<https://www.hhs.gov/web/governance/digital-strategy/it-policy-archive/hhs-policy-for-enterprise-architecture.html> and NIH EA requirements are located at: <https://ocio.nih.gov/PM/Pages/EPLC.aspx>.

- b. **System Documentation.** Contractors (and/or any subcontractors) must follow and adhere to HHS System Development Life Cycle requirements, at a minimum, for system development and provide system documentation at designated intervals (specifically, at the expiration of the contract) within the EPLC that require artifact review and approval.
 - c. **Sanitization of Government Files and Information.** As part of contract closeout and at expiration of the contract, the Contractor (and/or any subcontractor) must provide all required documentation in accordance with the NIH Media Sanitization and Disposal Policy to the CO and/or COR to certify that, at the government's direction, all electronic and paper records are appropriately disposed of and all devices and media are sanitized in accordance with NIST SP 800-88, *Guidelines for Media Sanitization*.
 - d. **Notification.** The Contractor (and/or any subcontractor) must notify the CO and/or COR and system ISSO within **fifteen days** before an employee stops working under this contract.
 - e. **Contractor Responsibilities upon Physical Completion of the Contract.** The Contractor (and/or any subcontractors) must return all government information and IT resources (i.e., government information in non-government-owned systems, media, and backup systems) acquired during the term of this contract to the CO and/or COR. Additionally, the Contractor must provide a certification that all government information has been properly sanitized and purged from Contractor-owned systems, including backup systems and media used during contract performance, in accordance with HHS and/or NIH policies.
 - f. The Contractor (and/or any subcontractor) must perform and document the actions identified in the NIH Employee Separation Checklist <https://ocio.nih.gov/aboutus/publicinfosecurity/acquisition/Documents/Emp-sep-checklist.pdf> when an employee terminates work under this contract within **2** days of the employee's exit from the contract. All documentation must be available to the CO and/or COR upon request.
9. Records Management and Retention
- a. The Contractor (and/or any subcontractor) must maintain all information in accordance with Executive Order 13556 -- Controlled Unclassified Information, National Archives and Records Administration (NARA) records retention policies and schedules and *HHS Policy for Records Management* and NIH policies and must not dispose of any records unless authorized by HHS/ NIH.
 - b. In the event that a contractor (and/or any subcontractor) accidentally disposes of or destroys a record without proper authorization, he/she must document and report the incident in accordance with HHS/ NIH policies.

SECTION B – PRIVACY ACT

It has been determined that this contract is subject to the Privacy Act of 1974, because this contract provides for the design, development, or operation of a system of records on individuals.

The System of Records Notice (SORNs) that are applicable to this contract are:

- [09-25-0108](#) Personnel: Guest Researchers, Special Volunteers, and Scientists Emeriti
 - SORN history: 67 FR 60742 at 60759 (9/26/02), *[83 FR 6591](#) (2/14/18)
- [09-25-0115](#) Administration: Curricula Vitae of Consultants and Clinical Investigators
 - SORN history: 67 FR 60742 at 60760 (9/26/02), *[83 FR 6591](#) (2/14/18)
- [09-25-0118](#) Contracts: Professional Services Contractors
 - SORN history: 67 FR 60742 at 60761 (9/26/02), *[83 FR 6591](#) (2/14/18)

The Contractor and any Subcontractor must follow disposition to be made of the Privacy Act records upon completion of contract performance shall be in accordance with Section C of the contract, and by direction of the Contracting Officer/Contracting Officer's Representative.

SECTION C – GOVERNMENT INFORMATION PROCESSED ON GOCO OR COCO SYSTEMS

SECURITY REQUIREMENTS FOR GOVERNMENT-OWNED/CONTRACTOR-OPERATED (GOCO) AND CONTRACTOR-OWNED/CONTRACTOR-OPERATED (COCO) RESOURCES

- a. **Federal Policies** - The Contractor (and/or any subcontractor) shall comply with applicable federal laws that include, but are not limited to, the HHS Information Security and Privacy Policy (IS2P); National Institutes of Health (NIH) Information Security (InfoSec) Policy Handbook; Federal Information Security Modernization Act (FISMA) of 2014, (44 U.S.C. 101); National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53, Security and Privacy Controls for Federal Information Systems and Organizations; Office of Management and Budget (OMB) Circular A-130, Managing Information as a Strategic Resource; and other applicable federal laws, regulations, NIST guidance, and Departmental policies.
- b. **Government Access for Security Assessment** - In addition to the Inspection Clause in the contract, the Contractor (and/or any subcontractor) shall afford the Government access to the Contractor's facilities, installations, operations, documentation, information systems, and personnel used in performance of this contract to the extent required to carry out a program of security assessment (to include vulnerability testing), investigation, and audit to safeguard against threats and hazards to the confidentiality, integrity, and availability of federal data or to the protection of information systems operated on behalf of HHS, including but are not limited to:

- i. At any tier handling or accessing information, consent to and allow the Government, or an independent third party working at the Government's direction, without notice at any time during a weekday during regular business hours Contractor local time, to access Contractor and subcontractor installations, facilities, infrastructure, data centers, equipment (including but not limited to all servers, computing devices, and portable media), operations, documentation (whether in electronic, paper, or other forms), databases, and personnel which are used in performance of the contract.

The Government includes but is not limited to the U.S. Department of Justice, U.S. Government Accountability Office, and the HHS Office of the Inspector General (OIG). The purpose of the access is to facilitate performance inspections and reviews, security and compliance audits, and law enforcement investigations. For security audits, the audit may include but not be limited to such items as buffer overflows, open ports, unnecessary services, lack of user input filtering, cross site scripting vulnerabilities, SQL injection vulnerabilities, and any other known vulnerabilities.

- ii. At any tier handling or accessing protected information, fully cooperate with all audits, inspections, investigations, forensic analysis, or other reviews or requirements needed to carry out requirements presented in applicable law or policy. Beyond providing access, full cooperation also includes, but is not limited to, disclosure to investigators of information sufficient to identify the nature and extent of any criminal or fraudulent activity and the individuals responsible for that activity. It includes timely and complete production of requested data, metadata, information, and records relevant to any inspection, audit, investigation, or review, and making employees of the Contractor available for interview by inspectors, auditors, and investigators upon request. Full cooperation also includes allowing the Government to make reproductions or copies of information and equipment, including, if necessary, collecting a machine or system image capture.
 - Segregate Government protected information and metadata on the handling of Government protected information from other information. Commingling of information is prohibited. Inspectors, auditors, and investigators will not be precluded from having access to the sought information if sought information is commingled with other information.
 - Cooperate with inspections, audits, investigations, and reviews.
- c. **End of Life Compliance** - The Contractor (and/or any subcontractor) must use Commercial off the Shelf (COTS) software or other software that is supported by the manufacturer. In addition, the COTS/other software need to be within one major version of the current version; deviation from this requirement will only be allowed via the HHS waiver process (approved by HHS CISO). The Contractor shall retire and/or upgrade all software/systems that have reached end-of-life in accordance with HHS End-of-Life Operating Systems, Software, and Applications Policy.
- d. **Desktops, Laptops, and Other Computing Devices Required for Use by the Contractor** - The Contractor (and/or any subcontractor) shall ensure that all IT equipment (e.g., laptops,

desktops, servers, routers, mobile devices, peripheral devices, etc.) used to process information on behalf of HHS are deployed and operated in accordance with approved security configurations and meet the following minimum requirements:

- i. Encrypt equipment and sensitive information stored and/or processed by such equipment in accordance with HHS and FIPS 140-3 encryption standards.
 - ii. Configure laptops and desktops in accordance with the latest applicable United States Government Configuration Baseline (USGCB), and HHS Minimum Security Configuration Standards;
 - iii. Maintain the latest operating system patch release and anti-virus software definitions within 15 days.
 - iv. Validate the configuration settings after hardware and software installation, operation, maintenance, update, and patching and ensure changes in hardware and software do not alter the approved configuration settings; and
 - v. Automate configuration settings and configuration management in accordance with HHS security policies, including but not limited to:
 - Configuring its systems to allow for periodic HHS vulnerability and security configuration assessment scanning; and
 - Using Security Content Automation Protocol (SCAP)-validated tools with USGCB Scanner capabilities to scan its systems at least on a monthly basis and report the results of these scans to the CO and/or COR, Project Officer, and any other applicable designated POC.
- e. **Rights to Data.** All contracts that require data to be produced, furnished, acquired, or used in meeting contract performance requirements, must contain terms that delineate the respective rights and obligations of the Government and the contractor regarding the use, reproduction, and disclosure of that data. Data rights clauses do not specify the type, quantity or quality of data that is to be delivered, but only the respective rights of the Government and the contractor regarding the use, disclosure, or reproduction of the data. Accordingly, the contract must specify the data to be delivered.
- f. **Information and Communications Technology (ICT) Cybersecurity Supply Chain Risk Management (C-SCRM) requirements.** The Contractor (and/or any subcontractor) must secure their ICT supply chain in compliance with *HHS Policy for Cyber Supply Chain Risk Management* and Public Law 115-232 § 889. At a minimum, they must implement the following:
- i. Develop rules for suppliers' development methods, techniques, or practices;
 - ii. Use of secondary market components;
 - iii. Prohibit counterfeit products;
 - iv. Dispose and/or retain elements such as components, data, or intellectual property securely;

- v. Ensure adequate supply of components;
- vi. Require external providers handling federal information or operating systems on behalf of the federal government to meet the same security and privacy requirements as federal agencies;
- vii. Require external providers to express security and privacy requirements (including the controls for systems processing, storing, or transmitting federal information) in contracts or other formal agreements;
- viii. Establish Service Level Agreements (SLAs), patching vehicles and disclosure requirements in the case of a security incident or new vulnerability being discovered; and
- ix. Ensure that the supplier applies same contractual requirements to any sub-contractors/suppliers that they involve in the provision of the product or service to the customer; and
- x. Prohibit the use of covered telecommunications and video surveillance equipment or services.

SECTION D – NON-COMMERCIAL AND OPEN SOURCE COMPUTER SOFTWARE PROCUREMENTS

1. The Contractor (and/or any subcontractor) must follow secure coding best practice requirements, as directed by the United States Computer Emergency Readiness Team (US-CERT) specified standards and the Open Web Application Security Project (OWASP) that will limit system software vulnerability exploits. The Contractor will be liable for malicious or defective code or failure to reduce risk.

SECTION E – INFORMATION TECHNOLOGY APPLICATION DESIGN, DEVELOPMENT, OR SUPPORT

1. The Contractor (and/or any subcontractor) must ensure IT applications designed and developed for end users (including mobile applications and software licenses) run in the standard user context without requiring elevated administrative privileges.
2. The Contractor (and/or any subcontractor) must follow secure coding best practice requirements, as directed by United States Computer Emergency Readiness Team (US-CERT) specified standards and the Open Web Application Security Project (OWASP), that will limit system software vulnerability exploits.
3. The Contractor (and/or any subcontractor) must ensure that computer software developed on behalf of HHS or tailored from an open-source product, is fully functional and operates correctly on systems configured in accordance with government policy and federal configuration standards. The contractor must test applicable products and versions with all relevant and current updates and patches updated prior to installing in the HHS environment. No sensitive data must be used during software testing.

4. The Contractor (and/or any subcontractor) must protect information that is deemed sensitive from unauthorized disclosure to persons, organizations or subcontractors who do not have a need to know the information. Information which, either alone or when compared with other reasonably-available information, is deemed sensitive or proprietary by HHS must be protected as instructed in accordance with the magnitude of the loss or harm that could result from inadvertent or deliberate disclosure, alteration, or destruction of the data. This language also applies to all subcontractors that are performing under this contract.

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